

CODE-GRAPH — KZTEK Labeling Studio

Tài liệu bản đồ codebase. Mọi coding agent PHẢI đọc file này TRƯỚC khi mở source.
Cập nhật ngay sau mỗi PR merge có thay đổi cấu trúc/API/schema.

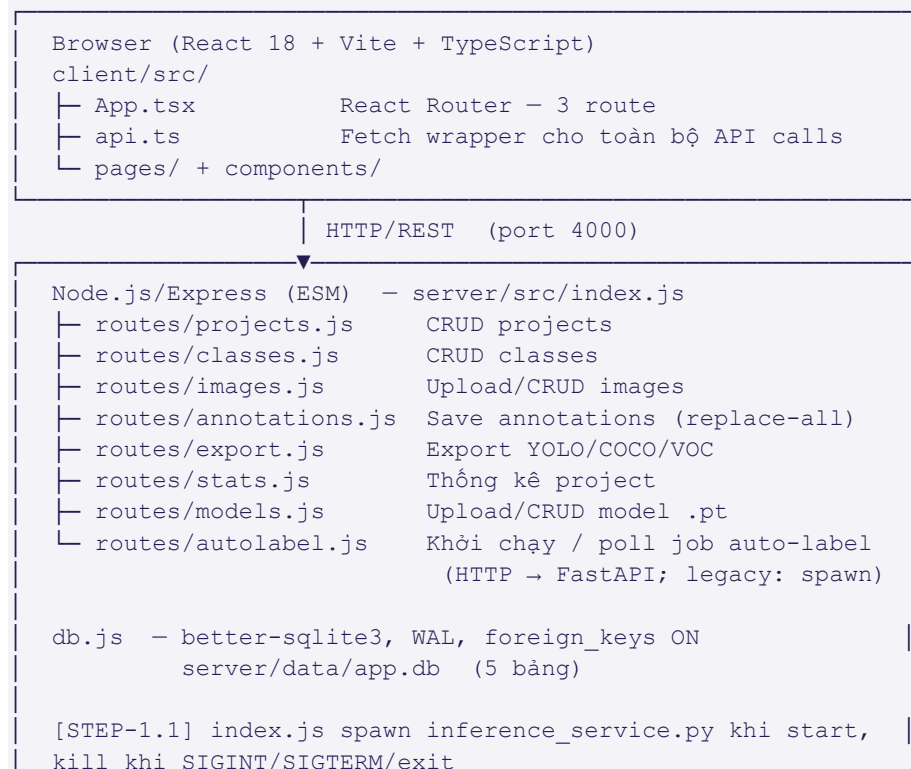
>

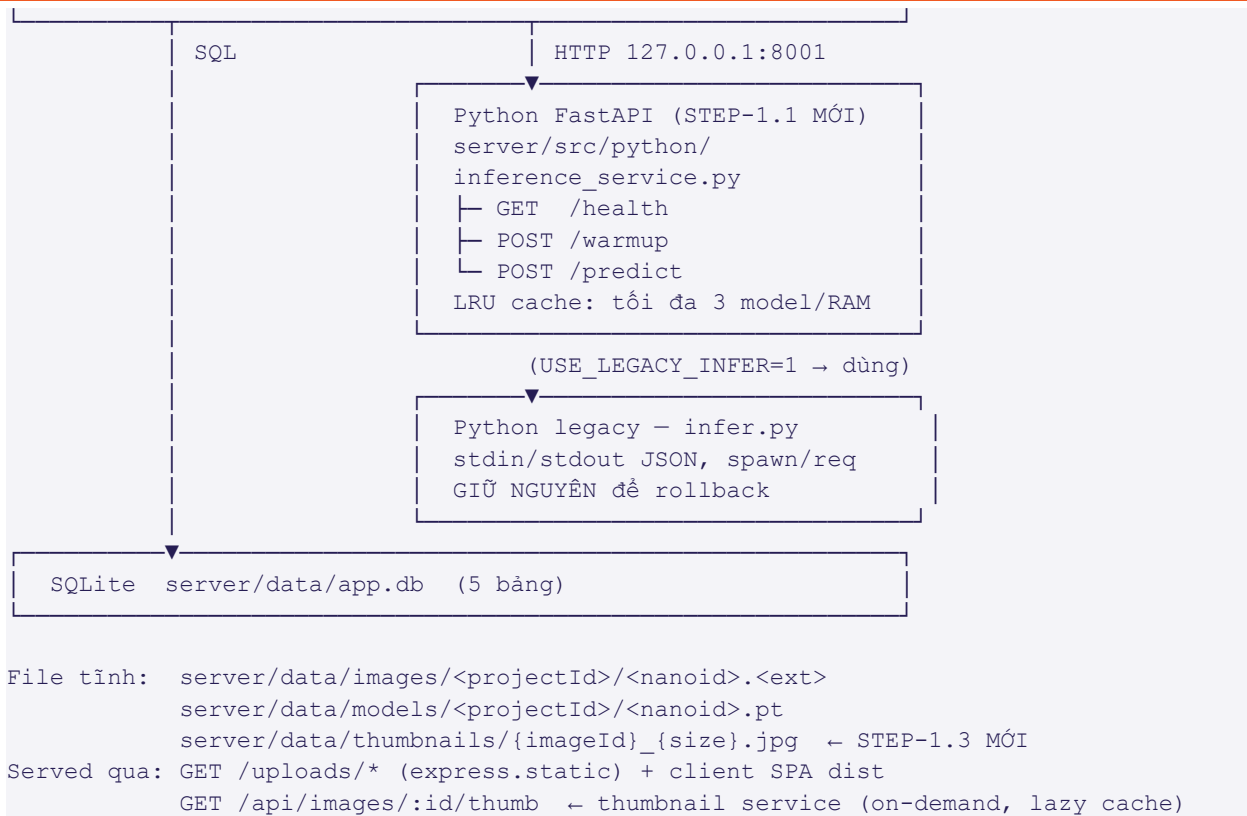
Last verified: 2026-08-05 | Cập nhật: senior-developer (STEP-4.1)

Mục lục

1. [Tổng quan kiến trúc](#1-tổng-quan-kiến-trúc)
2. [Cấu trúc thư mục](#2-cấu-trúc-thư-mục)
3. [Module map — Server](#3-module-map--server)
4. [Module map — Client](#4-module-map--client)
5. [Module map — Python Inference](#5-module-map--python-inference)
6. [DB Schema thực tế](#6-db-schema-thực-tế)
7. [API Endpoints](#7-api-endpoints)
8. [Dependency graph](#8-dependency-graph)
9. [Điểm sẽ bị ảnh hưởng bởi plan cải tiến](#9-điểm-sẽ-bị-ảnh-hưởng-bởi-plan-cải-tiến)
10. [Lịch sử cập nhật](#10-lịch-sử-cập-nhật)

1. Tổng quan kiến trúc





Port: 4000 (server + serve SPA dist + /uploads)

Inference port: 8001 (env `INFERENCE_PORT`, mặc định 8001)

DB file: `server/data/app.db` (tạo tự động nếu chưa có)

Python bin: env `PYTHON_BIN` hoặc `python` (mặc định)

Rollback: env `USE_LEGACY_INFER=1` → dùng `infer.py` spawn-per-request

2. Cấu trúc thư mục

Roboflow - Copy/	
├ server/	
│ └ src/	
│ │ └ index.js	← Entry point Express
│ │ └ db.js	← DB init, schema, migrations
│ │ └ routes/	
│ │ │ └ projects.js	
│ │ │ └ classes.js	
│ │ │ └ images.js	← multer, sharp, adm-zip; GET / trả thêm
└ thumbnail_url	
│ │ └ thumbnails.js	← thumbnail cache service (STEP-1.3 MỚI)
│ │ └ annotations.js	← STEP-3.2: ghi snapshot vào annotation_history sau
│ │ │ mỗi save	
│ │ │ └ history.js	← STEP-3.2 MỚI: GET /history + POST
│ │ │ └ /history/:v/revert	
│ │ │ └ activity.js	← STEP-3.3 MỚI: GET /activity (project activity log)
│ │ │ └ export.js	← archiver (yolo/coco/voc); STEP-3.3: gọi
└ logActivity	

```

├── stats.js
├── models.js
├── autolabel.js      ← HTTP → FastAPI (STEP-1.1); DB-backed jobs (STEP-
1.2)
├── jobs.js          ← CRUD /api/jobs (STEP-1.2 MỚI)
├── python/
│   └── inference_service.py ← FastAPI thường trực, LRU cache 3 model (STEP-
1.1 MỚI)
├── infer.py         ← stdin→stdout JSON batch inference (GIỮ LẠI,
rollback)
├── migrations/      ← SQL migration files (chỉ tham khảo; db.js inline
là nguồn sự thật)
│   └── 001_create_jobs.sql ← (STEP-1.2)
├── data/            ← RUNTIME (git-ignored)
│   ├── app.db
│   ├── images/<projectId>/
│   ├── thumbnails/   ← cache thumbnail (STEP-1.3 MỚI):
│   │   └── {imageId}_{size}.jpg
│   └── models/<projectId>/
│       └── package.json
├── client/
│   ├── src/
│   │   ├── main.tsx
│   │   ├── App.tsx    ← React Router setup
│   │   ├── api.ts     ← Tất cả fetch call
│   │   ├── types.ts   ← TS interfaces
│   │   ├── styles.css
│   │   ├── pages/
│   │   │   ├── ProjectsPage.tsx ← List/create/delete projects
│   │   │   ├── ProjectDetailPage.tsx ← Image grid, upload, filters
│   │   │   └── AnnotatorPage.tsx ← Canvas editor (bbox+quad), autosave
│   │   ├── components/
│   │   │   ├── Logo.tsx
│   │   │   ├── StatsPanel.tsx
│   │   │   ├── ExportModal.tsx
│   │   │   └── AutoLabelModal.tsx
│   │   ├── utils/
│   │   │   └── files.ts
│   └── package.json
├── code-graph/
│   ├── CODE-GRAPH.md ← file này
│   └── CODE-GRAPH.pdf
└── docs/plans/      ← Plan files
  
```

3. Module map — Server

3.1 `server/src/index.js` — Entry point

Thuộc tính	Giá trị	Confidence		
---	---	---		
Loại	Express app entry	CONFIRMED		
Port	`process.env.PORT` \	\	4000`	CONFIRMED
Middleware	cors(), express.json(limit 10mb), express.static(/uploads)	CONFIRMED		

SPA fallback	serve <code>client/dist/index.html</code> cho mọi non-API route	CONFIRMED		
Callers/Used-by	(entry point — không ai import)	CONFIRMED		
Imports	db.js (UPLOAD_DIR), tất cả 8 route files, node:child_process (spawn)	CONFIRMED		
[STEP-1.1] Inference lifecycle	<code>startInferenceService()</code> → spawn <code>inference_service.py</code> sau <code>app.listen()</code> . <code>stopInferenceService()</code> đăng ký qua <code>process.on('exit'/'SIGINT'/'SIGTERM')</code>	CONFIRMED		
[STEP-1.2] Jobs router	import + mount <code>routes/jobs.js</code> tại <code>/api/jobs</code>	CONFIRMED		
[STEP-1.3] Thumbnails router	import + mount <code>routes/thumbnails.js</code> tại <code>/api/images</code>	CONFIRMED		
[STEP-3.2] History router	import + mount <code>routes/history.js</code> tại <code>/api/images/:imageId</code>	CONFIRMED		
[STEP-3.3] Activity router	import + mount <code>routes/activity.js</code> tại <code>/api/projects/:projectId/activity</code>	CONFIRMED		
Env vars đọc	PORT, PYTHON_BIN, INFERENCE_PORT, USE_LEGACY_INFER	CONFIRMED		
Last verified	2026-08-05 (STEP-3.3)	-		

Route mounting:

<code>/api/projects</code>	→ <code>routes/projects.js</code>	
<code>/api/projects/:projectId/classes</code>	→ <code>routes/classes.js</code>	
<code>/api/projects/:projectId/images</code>	→ <code>routes/images.js</code>	
<code>/api/images/:imageId/annotations</code>	→ <code>routes/annotations.js</code>	
<code>/api/projects/:projectId/export</code>	→ <code>routes/export.js</code>	
<code>/api/projects/:projectId/stats</code>	→ <code>routes/stats.js</code>	
<code>/api/projects/:projectId/models</code>	→ <code>routes/models.js</code>	
<code>/api/projects/:projectId/auto-label</code>	→ <code>routes/autolabel.js</code>	
<code>/api/jobs</code>	→ <code>routes/jobs.js</code>	[STEP-1.2 MỚI]
<code>/api/images</code>	→ <code>routes/thumbnails.js</code>	[STEP-1.3 MỚI]
<code>/api/images/:imageId</code>	→ <code>routes/reviews.js</code>	[STEP-2.3 MỚI]
<code>/api/images/:imageId</code>	→ <code>routes/history.js</code>	[STEP-3.2 MỚI]
<code>/api/projects/:projectId/activity</code>	→ <code>routes/activity.js</code>	[STEP-3.3 MỚI]

3.2 `server/src/db.js` — Database layer

Thuộc tính	Giá trị	Confidence
---	---	---
Loại	Singleton DB module (ESM export)	CONFIRMED
Exports	<code>db</code> , <code>DATA_DIR</code> , <code>UPLOAD_DIR</code> , <code>MODEL_DIR</code> , <code>DB_PATH</code> , <code>HISTORY_MAX_VERSIONS</code> , <code>pruneAnnotationHistory()</code> , <code>logActivity()</code>	CONFIRMED
Callers/Used-by	<code>index.js</code> (UPLOAD_DIR), <code>projects.js</code> , <code>classes.js</code> , <code>images.js</code> , <code>annotations.js</code> , <code>export.js</code> , <code>stats.js</code> , <code>models.js</code> , <code>autolabel.js</code> , <code>jobs.js</code> , <code>history.js</code> ,	CONFIRMED

	activity.js	
DB engine	better-sqlite3, WAL mode, foreign_keys=ON	CONFIRMED
Schema init	CREATE TABLE IF NOT EXISTS — idempotent	CONFIRMED
Migrations inline	PRAGMA table_info + ALTER TABLE cho cột thiếu	CONFIRMED
[STEP-1.2] Startup cleanup	UPDATE jobs SET status='error' WHERE status IN ('running', 'pending') — mọi job dở dang khi restart đều bị đánh dấu error	CONFIRMED
[STEP-4.1] m007_detect_cache()	CREATE TABLE detect_cache + unique index (image_id, model_id) + 2 indexes; verify row count trước/sau	CONFIRMED
Last verified	2026-08-05 (STEP-4.1)	-

Inline migrations hiện có (chạy khi server khởi động):

1. `annotations.type` — nếu chưa có → ADD COLUMN type TEXT DEFAULT 'bbox'
2. `annotations.points` — nếu chưa có → ADD COLUMN points TEXT
3. `classes.hotkey` — nếu chưa có → ADD COLUMN hotkey TEXT
4. [STEP-1.2] Cleanup jobs — UPDATE running/pending → error (startup idempotent)
5. [STEP-2.2] `m003_add_users()` — CREATE TABLE users + seed admin + ADD COLUMN images.uploaded_by
6. [STEP-2.3] `m004_add_review_status()` — ALTER TABLE images (4 review cols) + index
7. [STEP-3.1] `m005_phase3_schema()` — ADD COLUMN annotations.version + CREATE TABLE annotation_history + activity_log; backup + verify row count
8. [STEP-3.5] `m006_add_image_done_columns()` — ALTER TABLE images (completed_at + completed_by) + index; verify row count
9. [STEP-4.1] `m007_detect_cache()` — CREATE TABLE detect_cache + 3 indexes; verify row count

3.3 `server/src/routes/projects.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router() (không mergeParams)	CONFIRMED
Imports	db	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects)	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	projects.js:9	List all projects (JOIN count images/labeled/classes)
POST	/	projects.js:20	Create project (auto-create 2 default classes)

GET	<code>/:id</code>	projects.js:33	Get project by id
PATCH	<code>/:id</code>	projects.js:39	Update name/description
DELETE	<code>/:id</code>	projects.js:48	Delete project (CASCADE images/classes/annotations)

3.4 `server/src/routes/classes.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/classes)	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	<code>/</code>	classes.js:13	List classes of project (ORDER BY sort_order)
POST	<code>/</code>	classes.js:19	Create class (auto sort_order)
PATCH	<code>/:classId</code>	classes.js:30	Update name/color/hotkey
DELETE	<code>/:classId</code>	classes.js:45	Delete class (CASCADE annotations)

Lưu ý: `hotkey` bị truncate còn 1 ký tự (`normalizeHotkey` function:L7).

3.5 `server/src/routes/images.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, UPLOAD_DIR, multer, sharp, adm-zip, nanoid	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/images)	CONFIRMED
Max file size	100MB / ảnh, 2GB / zip, 20000 ảnh / upload	CONFIRMED
Định dạng	JPG, PNG, WEBP, BMP	CONFIRMED
Last verified	2026-08-05 (STEP-3.4)	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	<code>/</code>	images.js:51	List images (kèm class_ids array + thumbnail_url — STEP-

GET	<code>/:imageId</code>	images.js:64	1.3) Get single image + annotations + annotationVersion (STEP-3.4)
POST	<code>/upload</code>	images.js:73	Upload nhiều file ảnh (multer.array)
POST	<code>/upload-zip</code>	images.js:95	Upload ZIP chứa ảnh
PATCH	<code>/:imageId</code>	images.js:143	Update split/status
DELETE	<code>/:imageId</code>	images.js:153	Delete image + xóa file vật lý

[STEP-3.4] GET `/:imageId` response nay trả thêm `annotationVersion: number = COALESCE(MAX(version), 0)` từ `annotation_history` của ảnh. Client dùng giá trị này làm `expectedVersion` khi gọi PUT annotations.

3.6 `server/src/routes/annotations.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, nanoid, pruneAnnotationHistory (từ db.js)	CONFIRMED
Callers/Used-by	index.js (mounted <code>/api/images/:imageId/annotations</code>)	CONFIRMED
Pattern	Replace-all: DELETE + bulk INSERT trong transaction; [STEP-3.2] INSERT snapshot vào <code>annotation_history</code> + prune; [STEP-3.4] Optimistic locking: đọc currentVersion từ annotation_history TRƯỚC transaction, check với expectedVersion từ client body	CONFIRMED
Last verified	2026-08-05 (STEP-3.4)	-

Method	Path	Mô tả	Response
---	---	---	---
PUT	<code>/</code>	Replace toàn bộ annotations; optimistic lock check; INSERT snapshot <code>annotation_history</code> (version=MAX+1); <code>pruneAnnotationHistory</code>	<code>200 {annotations: Annotation[], annotationVersion: number}</code> hoặc <code>409 {error: 'ANNOTATION_CONFLICT', serverVersion: N, message: string}</code>

[STEP-3.4] Optimistic locking:

- Client gửi `expectedVersion` trong body (integer — version client đang cầm từ lần GET gần nhất).

- Server đọc `COALESCE (MAX (version) , 0)` từ `annotation_history` TRƯỚC khi vào transaction.
- Nếu `expectedVersion !== currentVersion` → **409 Conflict** (không lưu gì).
- Nếu `expectedVersion` không gửi (undefined/null) → bỏ qua check (backward compat).
- Nếu pass → save bình thường, version tăng lên `currentVersion + 1`.

Version concept: "version của ảnh" = `MAX (version)` trong `annotation_history` (per-save, tăng mỗi lần PUT thành công). **KHÔNG phải** cột `annotations.version` per-row (cột đó chỉ là placeholder từ migration 3.1, chưa được dùng cho locking).

Quan trọng: Không có GET riêng — annotations được trả cùng `GET /images/:imageId` (images.js:64).

Depth-1 callers: index.js (mount), history.js (dùng chung DB nhưng không import annotations.js).

3.7 `server/src/routes/export.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, UPLOAD_DIR, archiver	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/export)	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả			
---	---	---	---			
GET	/split-preview	export.js:85	Preview phân chia train/valid/test trước export			
GET	/	export.js:114	Export ZIP (format=yolo\	coco\	voc, splitMode=manual\	auto)

Hỗ trợ: YOLO (bbox + OBB quad), COCO JSON, Pascal VOC XML. Stratified split logic (rarest class first).

3.8 `server/src/routes/stats.js`

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	stats.js:6	Tổng số ảnh, labeled, bySplit, perClass annotation count

3.9 `server/src/routes/models.js`

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	models.js:26	List models của project

POST	/upload	models.js:32	Upload file .pt (max 1GB)
DELETE	/:modelId	models.js:43	Delete model + xóa file .pt

3.10 `server/src/routes/autolabel.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, UPLOAD_DIR, MODEL_DIR, child_process.spawn, readline, node:path, nanoid	CONFIRMED
[STEP-1.2] Job storage	DB-backed: <code>createJobInDB()</code> khi POST; <code>syncJobToDB()</code> khi hoàn thành; <code>getJobFromDB()</code> khi GET fall-back. In-memory Map chỉ dùng trong quá trình inference đang chạy.	CONFIRMED
[STEP-1.1] Inference mode mặc định	HTTP fetch → <code>http://127.0.0.1:{INFERENCING_PORT}/predict</code> (FastAPI service thường trực)	CONFIRMED
[STEP-1.1] Rollback mode	<code>USE_LEGACY_INFER=1</code> → dùng spawn infer.py stdin/stdout cũ (giữ nguyên để rollback, KHÔNG có <code>detect_cache</code>)	CONFIRMED
Health gate	Trước mỗi job, gọi <code>GET /health</code> (timeout 3s); trả 503 nếu service chưa sẵn sàng	CONFIRMED
Batch size	32 ảnh/request tới FastAPI; timeout 5 phút/batch	CONFIRMED
[STEP-4.1] <code>detect_cache</code>	<code>CACHE_RAW_CONF=0.01</code> ; <code>getCachedDetections(imageId, modelId)</code> → cache hit: <code>filterRawBoxesByConf(conf)</code> → <code>saveDetections</code> ; cache miss: gọi inference với <code>CACHE_RAW_CONF</code> → <code>saveToCacheDetections()</code> → filter → save	CONFIRMED
Env vars đọc	<code>INFERENCING_PORT</code> (default 8001), <code>USE_LEGACY_INFER</code>	CONFIRMED
Last verified	2026-08-05 (STEP-4.1)	-

Method	Path	File:Line	Mô tả
---	---	---	---
POST	/	autolabel.js:~340	Khởi chạy job inference; tạo row DB ngay; returns <code>{jobId, total}</code> 202
GET	/:jobId	autolabel.js:~400	Poll trạng thái: memory Map → fall-back DB
DELETE	/cache	autolabel.js:~420	Xóa <code>detect_cache</code> entries theo project (+ filter <code>?imageId=&modelId=</code>). Auth required. Response: <code>{deleted: N}</code>

Cơ chế mới (STEP-1.1): `checkInferenceHealth()` → `fetch(INFERENCING_URL/predict, batch)` → parse response `{classes, results, errors}` → `saveDetectionsForImage()` → cập nhật job.

Cơ chế cache (STEP-4.1): Mỗi batch: tách `hits` (có cache) / `misses` (cần inference). Hits: load `raw_detections` từ `detect_cache` → `filterRawBoxesByConf(conf)` → `save`. Misses: gọi inference với `CACHE_RAW_CONF=0.01` → `saveToCacheDetections()` → `filterRawBoxesByConf(conf)` → `save`. Cho phép đổi `conf` threshold mà không detect lại model.

Cơ chế persist (STEP-1.2): `createJobInDB()` khi POST; HTTP mode `.then()/catch()` → `syncJobToDB()` → `jobs.delete()`. Legacy mode: `setInterval(500ms)` poll `job.status` → sync khi done/error.

Legacy (USE_LEGACY_INFER=1): `spawn(PYTHON_BIN, [INFER_SCRIPT])` → stdin JSON → readline stdout — giữ nguyên, không xóa. **KHÔNG dùng detect_cache trong legacy mode.**

Depth-1 callers: `index.js` (mount `/api/projects/:pid/auto-label`) — không thay đổi mount path.

3.12 `server/src/routes/thumbnails.js` — Thumbnail cache service (STEP-1.3 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	<code>Router()</code> (không <code>mergeParams</code>)	CONFIRMED
Imports	<code>db</code> , <code>UPLOAD_DIR</code> , <code>DATA_DIR</code> , <code>path</code> , <code>fs</code> , <code>sharp</code>	CONFIRMED
Callers/Used-by	<code>index.js</code> (mounted <code>/api/images</code>)	CONFIRMED
Cache dir	<code>DATA_DIR/thumbnails/</code> — tạo sẵn khi module load	CONFIRMED
Naming cache file	<code>{imageId}_{size}.jpg</code>	CONFIRMED
Size range	min 32, max 600, default 300	CONFIRMED
Fallback	resize lỗi → serve ảnh gốc (không trả 500)	CONFIRMED
Last verified	2026-08-04 (STEP-1.3)	-

Method	Path	Mô tả
---	---	---
GET	<code>/:id/thumb</code>	Lazy thumbnail: cache hit → serve; ảnh gốc nhỏ hơn size → serve gốc; else resize+cache+serve

Query params: `?size=300` (default 300, min 32, max 600)

Cache-Control: `public, max-age=86400` (1 ngày)

Exports: `THUMB_DIR` (dùng để test/cleanup)

3.11 `server/src/routes/jobs.js` — Job history CRUD (STEP-1.2 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	<code>Router()</code>	CONFIRMED
Imports	<code>db</code>	CONFIRMED
Callers/Used-by	<code>index.js</code> (mounted <code>/api/jobs</code>)	CONFIRMED
Last verified	2026-08-04 (STEP-1.2)	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	jobs.js:~22	List jobs by projectId (query param <code>?projectId=</code>) — ORDER BY created_at DESC
GET	<code>/:id</code>	jobs.js:~31	Get single job by id
DELETE	<code>/:id</code>	jobs.js:~37	Delete job record

Response shape (`normalizeJob`): `{id, projectId, status, total, done, created, failed, modelId, error, unmatchedClasses[], createdAt, updatedAt}`

3.14 `server/src/routes/history.js` — Annotation History (STEP-3.2 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	<code>Router({ mergeParams: true })</code>	CONFIRMED
Imports	<code>db, nanoid, pruneAnnotationHistory (db.js), requireRole (middleware/roles.js)</code>	CONFIRMED
Callers/Used-by	<code>index.js (mounted <code>/api/images/:imageId</code>)</code>	CONFIRMED
Last verified	2026-08-05 (STEP-3.2)	-

Method	Path	Role required	Mô tả
---	---	---	---
GET	<code>/history</code>	any authenticated	List history entries (id, version, actor_id, created_at, actor_name, actor_username) — KHÔNG kèm snapshot
POST	<code>/history/:version/revert</code>	reviewer, admin	Revert về version chỉ định: ghi đè annotations từ snapshot + INSERT history entry mới (audit) + <code>pruneAnnotationHistory</code>

Response GET `/history`: `[{id, version, actor_id, created_at, actor_name, actor_username}]` — sắp xếp DESC theo version.

Response POST `revert`: `{reverted_to_version: N, annotations: Annotation[]}`.

Role guard (AD-A5 Row 7): `annotator` → 403. Annotator được xem history (GET) nhưng không được revert.

Watch out: Sau revert, 1 history entry mới được tạo (version tăng tiếp) — audit trail liên tục, không bị gián đoạn.

3.15 `server/src/routes/activity.js` — Activity Log (STEP-3.3 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db (db.js)	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/activity)	CONFIRMED
Last verified	2026-08-05 (STEP-3.3)	-

Method	Path	Role required	Mô tả
---	---	---	---
GET	/	any authenticated	List activity log của project. LEFT JOIN users để lấy actor_name/actor_color. Phân trang: limit (mặc định 50, max 100) + offset. detail field trả về dưới dạng object (JSON.parse).

Response shape: `[[id, project_id, actor_id, action, detail: object|null, created_at, actor_name, actor_color]]` — ORDER BY created_at DESC.

Actions được ghi (bởi routes khác):

- `image_upload` — images.js (upload single + zip) → `{count, names, source?}`
- `image_delete` — images.js (DELETE) → `{image_id, filename}`
- `split_change` — images.js (PATCH) → `{image_id, from, to}`
- `export` — export.js → `{format, count}`

3.13 `server/src/routes/reviews.js` — Review workflow (STEP-2.3 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, requireRole (middleware/roles.js)	CONFIRMED
Callers/Used-by	index.js (mounted /api/images/:imageId)	CONFIRMED
Last verified	2026-08-04 (STEP-2.3)	-

Method	Path	Role required	Mô tả
---	---	---	---
POST	/submit-review	annotator, admin	draft\ rejected → in_review
POST	/approve	reviewer, admin	in_review → approved
POST	/reject	reviewer, admin	in_review →

			rejected (body: {comment})	
--	--	--	----------------------------	--

Lifecycle: draft → in_review → approved hoặc rejected → (submit lại) → in_review. Conflict trạng thái sai → 409 REVIEW_STATUS_CONFLICT.

4. Module map — Client

4.1 `client/src/App.tsx` — Router root

Route	Component	Mô tả
---	---	---
/	ProjectsPage	Danh sách projects
/projects/:projectId	ProjectDetailPage	Chi tiết project, image grid
/projects/:projectId/annotate/:imageId	AnnotatorPage	Canvas annotator

4.2 `client/src/api.ts` — API client

Thuộc tính	Giá trị	Confidence
---	---	---
Pattern	Thin fetch wrapper (<code>request<T></code>) — không có state	CONFIRMED
Callers/Used-by	ProjectsPage, ProjectDetailPage, AnnotatorPage	CONFIRMED
Auth	JWT Bearer + httpOnly cookie (STEP-2.2)	CONFIRMED
Last verified	2026-08-05 (STEP-3.4)	-

Hàm public:

```
listProjects, createProject, getProject, deleteProject
listClasses, createClass, updateClass, deleteClass
listImages, getImage, uploadImages, uploadZip, updateImage, deleteImage
saveAnnotations(imageId, annotations, expectedVersion?) ← [STEP-3.4] thêm tham số expectedVersion
  → trả { annotations: Annotation[], annotationVersion: number } (thay vì Annotation[] cũ)
  → throw Error('ANNOTATION_CONFLICT') nếu server trả 409
exportUrl, getSplitPreview
getStats
listModels, uploadModel, deleteModel
startAutoLabel, getAutoLabelJob
submitReview, approveReview, rejectReview
```

4.3 `client/src/types.ts` — TypeScript types

Các interface chính:

- **Project** — id, name, description, created_at, image_count, labeled_count, class_count
- **ClassLabel** — id, project_id, name, color, sort_order, hotkey
- **ImageItem** — id, project_id, filename, original_name, width, height, split, status, created_at, class_ids[], thumbnail_url? (STEP-1.3), review_status?, review_comment?, reviewed_by?, reviewed_at? (STEP-2.3)
- **Annotation** — id, image_id, class_id, x, y, w, h, type, points
- **ImageWithAnnotations** — ImageItem + annotations[] + **annotationVersion: number** (STEP-3.4 — version dùng cho optimistic locking)
- **ModelInfo** — id, project_id, filename, original_name, created_at
- **AutoLabelJob** — status, total, done, created, failed, error, unmatchedClasses[]

4.4 `client/src/pages/AnnotatorPage.tsx` — Canvas editor

Thuộc tính	Giá trị	Confidence
---	---	---
Callers/Used-by	App.tsx (route /projects/:pid/annotate/:iid)	CONFIRMED
API calls	api.listClasses, api.listImages, api.getImage, api.saveAnnotations, api.submitReview, api.approveReview, api.rejectReview	CONFIRMED
Autosave	debounce 600ms sau mỗi thay đổi box	CONFIRMED
Tools	bbox (draw rect), quad (4-point OBB)	CONFIRMED
Hotkeys	Arrow ← → navigate, Delete xóa box	CONFIRMED
State: boxes	Box[] (local) — sync lên server khi dirty	CONFIRMED
State: undo	KHÔNG có undo/redo hiện tại	CONFIRMED
[STEP-3.4] annotationVersionRef	useRef<number>(0) — lưu version annotation hiện tại (từ img.annotationVersion khi load). Dùng làm expectedVersion khi gọi saveAnnotations(). Cập nhật thành result.annotationVersion sau mỗi save thành công.	CONFIRMED
[STEP-3.4] 409 handling	scheduleSave catch Error('ANNOTATION_CONFLICT') → alert + reload image + cập nhật annotationVersionRef. Lỗi khác → trạng thái dirty để user retry.	CONFIRMED
Last verified	2026-08-05 (STEP-3.4)	-

4.5 `client/src/pages/ProjectDetailPage.tsx` — Image grid

API calls	listClasses, listImages, uploadImages, uploadZip, updateImage, deleteImage
-----------	--

---	---
Features	Filter by status/split/class/search, pagination (page×pageSize), drag-drop upload
Modals	StatsPanel, ExportModal, AutoLabelModal

4.6 Components

Component	Callers	Mô tả
---	---	---
Logo.tsx	App.tsx	KZTEK logo SVG inline
StatsPanel.tsx	ProjectDetailPage	Chart/bảng stats project
ExportModal.tsx	ProjectDetailPage	UI chọn format/split, trigger export
AutoLabelModal.tsx	ProjectDetailPage	Chọn model, confidence, scope; poll job progress

5. Module map — Python Inference

5.1 `server/src/python/infer.py` — Legacy (GIỮ LẠI để rollback)

Thuộc tính	Giá trị	Confidence
---	---	---
Protocol	stdin: JSON payload → stdout: streaming NDJSON (1 line/ảnh) + {"done":true}	CONFIRMED
Caller	autolabel.js runInference() (khi USE_LEGACY_INFER=1)	CONFIRMED
Dependencies	ultralytics (import lazy bên trong main())	CONFIRMED
Model load	YOLO(model_path) — load từ disk mỗi lần spawn	CONFIRMED
Output bbox	{class_index, type:"bbox", x, y, w, h} (pixel coords)	CONFIRMED
Output quad	{class_index, type:"quad", points:[{x,y}×4]} (OBB)	CONFIRMED
Error handling	Per-image try/except → {image_id, error, classes}	CONFIRMED
Status	GIỮ NGUYÊN — rollback bằng USE_LEGACY_INFER=1	CONFIRMED
Last verified	2026-08-04	-

Input format (stdin):

```
{
  "model_path": "/abs/path/model.pt",
```

```
"conf": 0.25,
"images": [{"id": "nanoid", "path": "/abs/path/img.jpg"}, ...]
}
```

5.2 `server/src/python/inference\_service.py` — FastAPI thường trực (STEP-1.1 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Framework	FastAPI + uvicorn	CONFIRMED
Port	127.0.0.1:{INFERENCE_PORT} (default 8001)	CONFIRMED
LRU cache	LRUModelCache(max_size=3) — tối đa 3 model giữ trong RAM	CONFIRMED
Caller	index.js startInferenceService() — spawn khi Node start	CONFIRMED
Dependencies	fastapi, uvicorn, ultralytics (lazy import bên trong <code>load_model</code>)	CONFIRMED
Lifecycle	index.js spawn/kill; không auto- restart khi crash (STEP log <code>Service stopped</code>)	CONFIRMED
Env vars	INFERENCE_PORT	CONFIRMED
Last verified	2026-08-04 (STEP-1.1)	-

Endpoints:

Method	Path	Mô tả
---	---	---
GET	/health	Trả {status, loaded_models[], uptime_s}
POST	/warmup	Preload model vào LRU cache. Body: {model_path}
POST	/predict	Batch inference. Body: {model_path, conf, iou, images:[{id,path}]}

Response `/predict`:

```
{
  "classes": ["class_name_0", ...],
  "results": [{"image_id": "nanoid", "detections":
[{"class_index":0,"type":"bbox","x":...,"y":...,"w":...,"h":...}]},
  "errors": [{"image_id": "nanoid", "error": "message"}]
}
```

Output bbox: {class_index, conf, type:"bbox", x, y, w, h} — **STEP-4.1:** thêm conf (float 0-1) để Node.js lưu raw detections vào detect_cache

Output quad (OBB): {class_index, conf, type:"quad", points:[{x,y}×4]} — **STEP-4.1:** thêm conf

Error handling: Per-image try/except → gộp vào errors[] (không crash toàn batch)

Backward compat: `saveDetectionsForImage()` trong Node.js không đọc `conf` (ignore) — KHÔNG break existing behavior. `conf` chỉ được dùng bởi `filterRawBoxesByConf()` khi lấy từ `detect_cache`.

6. DB Schema thực tế

Nguồn: `server/src/db.js` — đọc trực tiếp CREATE TABLE statements. Confidence: CONFIRMED toàn bộ.

Bảng ``projects``

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
name	TEXT	NOT NULL
description	TEXT	DEFAULT ""
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Bảng ``classes``

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
name	TEXT	NOT NULL
color	TEXT	NOT NULL
sort_order	INTEGER	NOT NULL DEFAULT 0
hotkey	TEXT	nullable (migration nếu chưa có)

Index: `idx_classes_project` ON `classes(project_id)`

Bảng ``images``

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
filename	TEXT	NOT NULL (tên file trên disk, nanoid+ext)
original_name	TEXT	NOT NULL (tên gốc khi upload)
width	INTEGER	NOT NULL
height	INTEGER	NOT NULL
split	TEXT	NOT NULL DEFAULT 'train' (train/valid/test)
status	TEXT	NOT NULL DEFAULT 'unlabeled' (unlabeled/labeled)
created_at	TEXT	NOT NULL DEFAULT

review_status	TEXT	datetime('now')	
		NOT NULL DEFAULT 'draft' — STEP-2.3 MỚI	rejected)
review_comment	TEXT	(draft→in_review→approved\ nullable — STEP-2.3 MỚI (lý do reject)	
reviewed_by	INTEGER	FK→users(id), nullable — STEP-2.3 MỚI	
reviewed_at	TEXT	nullable — STEP-2.3 MỚI	
completed_at	TEXT	nullable — STEP-3.5 MỚI (ISO datetime, NULL = chưa done)	
completed_by	INTEGER	FK→users(id), nullable — STEP-3.5 MỚI (người đã mark done)	

Index: `idx_images_project` ON `images(project_id)`, `idx_images_review_status` ON `images(review_status)` (STEP-2.3), `idx_images_completed_at` ON `images(completed_at)` (STEP-3.5)

ADR (done status v3): dùng 2 cột riêng `completed_at/completed_by`, KHÔNG mở rộng enum `status` — tách biệt "có annotation" (`status=labeled`) và "annotator xác nhận xong" (`completed_at` NOT NULL).

Gate submit-review (STEP-3.5): `submit-review` PHẢI có `completed_at` IS NOT NULL — trả 409 `IMAGE_NOT_COMPLETED` nếu chưa done.

Bảng `annotations`

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
image_id	TEXT	NOT NULL, FK→images(id) ON DELETE CASCADE
class_id	TEXT	NOT NULL, FK→classes(id) ON DELETE CASCADE
x	REAL	NOT NULL (pixel, top-left)
y	REAL	NOT NULL (pixel, top-left)
w	REAL	NOT NULL (pixel width)
h	REAL	NOT NULL (pixel height)
type	TEXT	NOT NULL DEFAULT 'bbox' (bbox/quad) — migration nếu chưa có
points	TEXT	nullable, JSON string <code>[[{x,y}×4]]</code> cho type=quad — migration nếu chưa có
created_at	TEXT	NOT NULL DEFAULT datetime('now')
version	INTEGER	NOT NULL DEFAULT 0 — STEP-3.1 MỚI (optimistic locking, dùng ở STEP-3.4)

Index: `idx_annotations_image` ON `annotations(image_id)`

Bảng `annotation\_history` — STEP-3.1 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	INTEGER	PRIMARY KEY AUTOINCREMENT
image_id	TEXT	NOT NULL, FK→images(id) ON DELETE CASCADE
version	INTEGER	NOT NULL
snapshot	TEXT	NOT NULL — JSON toàn bộ annotations của ảnh tại thời điểm save (SNAPSHOT, không diff — ADR AD-5)
actor_id	INTEGER	FK→users(id), nullable
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_ann_history_image, idx_ann_history_img_ver(image_id, version DESC)`. **Retention:** tối đa 200 version/ảnh qua `pruneAnnotationHistory(imageId)` (export từ `db.js`, gọi bởi STEP-3.2).

Bảng `activity\_log` — STEP-3.1 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	INTEGER	PRIMARY KEY AUTOINCREMENT
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
actor_id	INTEGER	FK→users(id), nullable
action	TEXT	NOT NULL (VD: 'upload', 'export', 'split_change')
detail	TEXT	nullable, JSON tùy ý
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_activity_log_project, idx_activity_log_action`. Route ghi log: STEP-3.3 (bảng đã tạo, chưa có route ghi ở STEP-3.1).

⚠ **Lưu ý:** `actor_id` không có `ON DELETE CASCADE/SET NULL` (mặc định NO ACTION) — không thể hard-delete user đã có history/activity_log khi `foreign_keys=ON`. Dùng `is_active=0` (soft-delete, đã có từ STEP-2.2) thay vì xóa thật.

Bảng `models`

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
filename	TEXT	NOT NULL (tên file .pt trên disk)
original_name	TEXT	NOT NULL (tên gốc khi upload)
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_models_project` ON `models(project_id)`

Bảng `detect\_cache` — STEP-4.1 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	INTEGER	PRIMARY KEY AUTOINCREMENT
image_id	TEXT	NOT NULL, FK→images(id) ON DELETE CASCADE
model_id	TEXT	NOT NULL, FK→models(id) ON DELETE CASCADE
raw_detections	TEXT	NOT NULL — JSON: {classes:string[], boxes:[{class_index, conf, x, y, w, h, type, points?}]} — conf gốc, KHÔNG áp threshold
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_detect_cache_img_model` UNIQUE ON `detect_cache(image_id, model_id)` — 1 entry/cặp ảnh-model; `idx_detect_cache_image`, `idx_detect_cache_model`

Cascade: ON DELETE CASCADE cả `image_id` lẫn `model_id` → cache tự xóa khi ảnh hoặc model bị xóa.

Thiết kế: Cache theo (`image_id`, `model_id`) — không cache theo `conf/iou`. `raw_detections` lưu ở `CACHE_RAW_CONF=0.01` để cho phép đổi `conf` user mà không detect lại. `IoU=0.45` (default inference service) — NMS đã áp tại inference, không thể thay đổi từ cache.

INSERT OR REPLACE: `saveToCacheDetections()` dùng UPSERT — detect lại cùng ảnh+model sẽ cập nhật cache.

Bảng `jobs` — STEP-1.2 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL (no FK — không CASCADE khi xóa project, job history giữ lại)
status	TEXT	NOT NULL DEFAULT 'pending' (pending/running/done/error)
total_images	INTEGER	NOT NULL DEFAULT 0
processed	INTEGER	NOT NULL DEFAULT 0 (ảnh đã xử lý xong)
created_annotations	INTEGER	NOT NULL DEFAULT 0
failed	INTEGER	NOT NULL DEFAULT 0
model_id	TEXT	nullable
error_msg	TEXT	nullable
unmatched_classes	TEXT	NOT NULL DEFAULT '[]' (JSON array tên class không map được)
created_at	TEXT	NOT NULL DEFAULT datetime('now')
updated_at	TEXT	NOT NULL DEFAULT datetime('now')

Indexes: `idx_jobs_project` ON `jobs(project_id)`, `idx_jobs_status` ON `jobs(status)`

Startup behavior: Khi server khởi động, mọi row có `status` IN ('running', 'pending') → cập nhật thành 'error' với `error_msg='Server restarted while job was in progress'`.

7. API Endpoints

Tất cả CONFIRMED từ source (index.js route mounting + từng route file).

Projects

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects	projects.js:9	Project[] (kèm counts)
POST	/api/projects	projects.js:20	Project (201)
GET	/api/projects/:id	projects.js:33	Project
PATCH	/api/projects/:id	projects.js:39	Project
DELETE	/api/projects/:id	projects.js:48	204

Classes

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/classes	classes.js:13	ClassLabel[]
POST	/api/projects/:pid/classes	classes.js:19	ClassLabel (201)
PATCH	/api/projects/:pid/classes/:cid	classes.js:30	ClassLabel
DELETE	/api/projects/:pid/classes/:cid	classes.js:45	204

Images

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/images	images.js:51	ImageItem[]
GET	/api/projects/:pid/images/:iid	images.js:64	ImageWithAnnotations
POST	/api/projects/:pid/images/upload	images.js:73	ImageItem[] (201)
POST	/api/projects/:pid/images/upload-zip	images.js:95	{created, skipped} (201)
PATCH	/api/projects/:pid/images/:iid	images.js:143	ImageItem
POST	/api/projects/:pid/images/:iid/mark-done	images.js:~200	ImageItem (completed_at, completed_by set) — STEP-3.5 MỚI
DELETE	/api/projects/:pid/images/:iid/mark-done	images.js:~220	ImageItem (completed_at=null, completed_by=null) — STEP-3.5 MỚI; annotator chỉ bỏ của mình (403 nếu người khác)
DELETE	/api/projects/:pid/images/:iid	images.js:~253	204

Annotations

Method	Path	File:Line	Response
---	---	---	---
PUT	/api/images/:iid/annotations	annotations.js:14	200 {annotations:

			Annotation[], annotationVersion: number} — [STEP-3.4] hoặc 409 {error:'ANNOTATION_CONFLICT', serverVersion:N, message:str} khi conflict. Body: {annotations, expectedVersion?}.
--	--	--	---

Annotation History (STEP-3.2 MỚI)

Method	Path	File:Line	Role	Response
---	---	---	---	---
GET	/api/images/:iid/history	history.js:26	any authenticated	<pre>[{id, version, actor_id, created_at, actor_name, actor_username}] — không có snapshot</pre>
POST	/api/images/:iid/history/:version/revert	history.js:47	reviewer, admin	<pre>{reverted_to_version: N, annotations: Annotation[]}</pre>

Export

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/export/split-preview	export.js:85	SplitPreview JSON
GET	/api/projects/:pid/export	export.js:114	ZIP stream (attachment)

Query params export: `format=yolo\|coco\|voc, splitMode>manual\|auto, trainRatio=0.8`

Stats

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/stats	stats.js:6	ProjectStats JSON

Models

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/models	models.js:26	ModelInfo[]
POST	/api/projects/:pid/models/upload	models.js:32	ModelInfo (201)
DELETE	/api/projects/:pid/models/:mid	models.js:43	204

Auto-label

Method	Path	File:Line	Response
---	---	---	---

POST	/api/projects/:pid/autolabel	autolabel.js:~340	{jobId, total} (202)
GET	/api/projects/:pid/autolabel/:jid	autolabel.js:~400	AutoLabelJob (memory → DB fallback)
DELETE	/api/projects/:pid/autolabel/cache	autolabel.js:~420	{deleted: N} — xóa detect_cache entries của project; filter: ?imageId=, ?modelId= (tùy chọn); auth required (mọi role); 404 nếu project không tồn tại — STEP-4.1 MỚI

Jobs (STEP-1.2 MỚI)

Method	Path	File:Line	Response
---	---	---	---
GET	/api/jobs?projectId=:pid	jobs.js:~22	JobRecord[]
GET	/api/jobs/:id	jobs.js:~31	JobRecord
DELETE	/api/jobs/:id	jobs.js:~37	{ok:true}

Thumbnails (STEP-1.3 MỚI)

Method	Path	File:Line	Response
---	---	---	---
GET	/api/images/:id/thumb	thumbnails.js:29	JPEG binary (thumbnail or original) — 200; 404 nếu image/file không tồn tại

Query params: ?size=300 (default 300px, min 32, max 600)

Cache-Control: public, max-age=86400

8. Dependency graph

Server dependencies (package.json — CONFIRMED)

Package	Version	Dùng ở
---	---	---
express	^4.21.0	index.js, tất cả routes
better-sqlite3	^9.6.0	db.js
cors	^2.8.5	index.js
multer	^1.4.5-lts.1	images.js, models.js
sharp	^0.32.6	images.js (metadata width/height, resize)
adm-zip	^0.6.0	images.js (unzip upload)
archiver	^7.0.1	export.js (tạo zip export)
nanoid	^5.0.7	projects, classes, images, annotations, models, autolabel

Client dependencies (package.json — CONFIRMED)

Package	Version	Dùng ở
---	---	---
react	^18.3.1	tất cả components/pages
react-dom	^18.3.1	main.tsx
react-router-dom	^6.26.2	App.tsx, các pages
vite	^5.4.6	dev/build tool
typescript	^5.6.2	type checking

Python dependencies

Package	Dùng ở	Confidence
---	---	---
ultralytics	infer.py, inference_service.py (YOLO model, predict — lazy import)	CONFIRMED
fastapi	inference_service.py (FastAPI app, endpoints, Pydantic schemas)	CONFIRMED
uvicorn	inference_service.py (ASGI server, entrypoint)	CONFIRMED

9. Điểm sẽ bị ảnh hưởng bởi plan cải tiến

Mục này là "blast radius map" cho 23 bước trong PLAN-MASTER.md.
Confidence: CONFIRMED từ phân tích source + plan.

Phase 1.1 — Inference service FastAPI (thường trực) — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-1.1):

- `server/src/routes/autolabel.js` — Refactor: thêm `runInferenceHTTP()` gọi FastAPI, giữ `runInference()` cũ cho rollback (USE_LEGACY_INFER=1). Health gate trước mỗi job → 503 khi service down.
- `server/src/index.js` — Thêm `startInferenceService()` / `stopInferenceService()` lifecycle.
- `server/src/python/inference_service.py` — FastAPI app mới, LRU cache 3 model.
- `server/src/python/infer.py` — GIỮ NGUYÊN (rollback).
- `server/README.md` — Thêm mục "Chạy inference service", env vars, cấu trúc Python inference.

Depth-1 callers: `index.js` (mount `autolabelRouter`) — không đổi mount path.

Watch out: `client/api.ts` `startAutoLabel` + `getAutoLabelJob` không đổi (same REST interface).

Phase 1.2 — Persist bảng `jobs` — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-1.2):

- `server/src/db.js` — thêm `CREATE TABLE IF NOT EXISTS jobs (...)` + 2 indexes + startup cleanup (UPDATE running/pending → error)

- `server/src/routes/autolabel.js` — thêm `createJobInDB`, `syncJobToDB`, `getJobFromDB`; POST handler tạo DB row + sync khi hoàn thành; GET handler fall-back DB
- `server/src/routes/jobs.js` — **MỚI**: GET/GET/:id/DELETE /api/jobs
- `server/src/index.js` — import + mount jobsRouter tại `/api/jobs`
- `server/migrations/001_create_jobs.sql` — **MỚI**: migration SQL (tham khảo)

Depth-1 callers: `client/src/api.ts` (`getAutoLabelJob`) — REST interface giữ nguyên, không cần sửa client.

Watch out: `jobs` table không có FK tới `projects` — khi xóa project, jobs không bị cascade. Thiết kế có chủ ý (giữ job history). Phase sau nếu cần cleanup → `DELETE FROM jobs WHERE project_id=?` trong `routes/projects.js` khi DELETE project.

Phase 1.3 — Thumbnail service — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-1.3):

- `server/src/routes/thumbnails.js` — **MỚI**: lazy thumbnail endpoint GET `/api/images/:id/thumb`, cache `DATA_DIR/thumbnails/{id}_{size}.jpg`
- `server/src/index.js` — import + mount thumbnailsRouter tại `/api/images`
- `server/src/routes/images.js` — GET / thêm field `thumbnail_url` vào response
- `client/src/types.ts` — thêm `thumbnail_url?: string` vào `ImageItem`
- `client/src/pages/ProjectDetailPage.tsx` — image grid dùng `img.thumbnail_url` thay `/uploads/...`

Depth-1 callers:

- `client/src/pages/ProjectDetailPage.tsx` → dùng `img.thumbnail_url` (thêm)
- `client/src/pages/AnnotatorPage.tsx` → không thay đổi (dùng `getImage` riêng, không qua list)

Watch out: `THUMB_DIR` tạo khi module load — nếu `DATA_DIR` không ghi được sẽ crash lúc server start. Cần đảm bảo `server/data/` có write permission.

Phase 2.2 — Auth implementation — ✔ HOÀN THÀNH

Files bị ảnh hưởng:

- `server/src/db.js` — bảng `users` (INTEGER AUTOINCREMENT id, username, password_hash, display_name, role, color, created_at, is_active)
- `server/src/index.js` — mount `auth.js/users.js`, middleware auth
- Tạo mới: `server/src/routes/auth.js`, `server/src/routes/users.js`, `server/src/middleware/`, `server/src/lib/`
- `client/src/api.ts`, `client/src/types.ts`, `client/src/App.tsx`, `client/src/pages/LoginPage.tsx`
- Test: `tests/auth.test.js` — 87 passed theo ma trận AD-A5 (2 điều kiện CTO #A1/#A2 đạt)

Phase 2.3 — Review workflow — ✔ HOÀN THÀNH

Files bị ảnh hưởng:

- `server/src/db.js` — `m004_add_review_status()`: 4 cột mới trên `images` (`review_status`, `review_comment`, `reviewed_by`, `reviewed_at`) + index

- Tạo mới: `server/src/routes/reviews.js` (xem §3.13)
- `server/src/index.js` — mount `reviewsRouter` tại `/api/images/:imageId`
- Test: `tests/auth.test.js` Row 8 — 8 test case pass (submit-review, approve, reject, comment, role guard, 401)

Đã xử lý ở STEP-3.5: `submit-review` trong `reviews.js` đã check `completed_at IS NOT NULL` — trả 409 `IMAGE_NOT_COMPLETED` nếu chưa done.

Phase 3.1 — DB Migration (ĐỤNG SCHEMA HIỆN CÓ) — ✔ HOÀN THÀNH

Files bị ảnh hưởng:

- `server/src/db.js` — `m005_phase3_schema()`: cột `annotations.version` DEFAULT 0, bảng `annotation_history` (snapshot), `activity_log`; auto-backup `.bak-{ts}` (giữ 5 bản) + verify row count trước/sau (throw nếu mất dữ liệu — CTO condition #1); export `pruneAnnotationHistory()`.
- Chưa đụng `server/src/routes/annotations.js` — route UPDATE version + INSERT history là việc của STEP-3.2, bước này CHỈ tạo schema.

Kết quả verify row count (thực tế đã chạy): projects 2→2, classes 4→4, images 1→1, annotations 0→0, models 0→0, jobs 0→0, users 1→1 — không mất dữ liệu ở bảng nào.

Depth-1 callers sẽ break nếu STEP-3.2 sai: tất cả code đọc/ghi bảng `annotations` (`annotations.js`, `images.js:64`, `export.js`, `stats.js`, `autolabel.js`) — CHƯA đổi ở bước này, cần re-verify khi STEP-3.2 sửa `annotations.js`.

Watch out: `annotation_history.snapshot` là SNAPSHOT JSON (không diff) — STEP-3.2 revert = ghi đè thẳng, không merge. `actor_id` không có ON DELETE CASCADE — không hard-delete user có history.

Phase 3.2 — Annotation history + revert — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-3.2):

- `server/src/routes/annotations.js` — import `pruneAnnotationHistory`; PUT handler: trong transaction thêm `COALESCE(MAX(version), 0) + 1` → INSERT snapshot → gọi `pruneAnnotationHistory(imageId)` ngoài transaction.
- `server/src/routes/history.js` — **MỚI:** GET `/history` (list, no snapshot) + POST `/history/:version/revert` (reviewer/admin, AD-A5 Row 7).
- `server/src/index.js` — import + mount `historyRouter` tại `/api/images/:imageId`.
- `tests/auth.test.js` — Row 7 (3 skip → 6 test thật): 93 passed, 0 failed, 0 skipped.

Depth-1 callers của annotations.js: không thay đổi interface PUT (input/output giống trước), chỉ thêm side-effect ghi history.

Depth-1 của history.js: `index.js` (mount) — client chưa gọi (frontend history panel là việc của phase UI sau).

Watch out cho STEP-3.4: Cần sửa `annotations.js` PUT handler — thêm `req.body.version` conflict check. STEP-3.3 KHÔNG sửa `annotations.js` (chỉ sửa `images.js/export.js`). Không xung đột.

Phase 3.3 — Activity log cấp project — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-3.3):

- `server/src/db.js` — thêm `export logActivity(projectId, actorId, action, detail): INSERT non-blocking vào activity_log.`
- `server/src/routes/activity.js` — **MỚI:** `GET /api/projects/:projectId/activity` (phân trang, LEFT JOIN users, detail dưới dạng object).
- `server/src/routes/images.js` — gọi `logActivity` sau: `upload (single + zip) → image_upload`, `PATCH` khi `split` đổi `→ split_change`, `DELETE → image_delete`.
- `server/src/routes/export.js` — gọi `logActivity` trước `archive.finalize()` `→ export`.
- `server/src/index.js` — mount `activityRouter` tại `/api/projects/:projectId/activity`.
- `tests/auth.test.js` — Row 17: verify `image_upload/split_change/image_delete` log + all roles 200 + unauth 401.

Kết quả test: 101 passed, 0 failed, 0 skipped. Commit: a0f01dc.

Phase 3.4 — Optimistic locking annotation — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-3.4):

- `server/src/routes/annotations.js` — PUT handler: đọc `currentVersion = COALESCE(MAX(version), 0)` từ `annotation_history` TRƯỚC transaction; if `expectedVersion` mismatch `→ 409`; response đổi thành `{annotations, annotationVersion}`.
- `server/src/routes/images.js` — `GET /:imageId` thêm `annotationVersion` vào response (query `annotation_history`).
- `client/src/types.ts` — `ImageWithAnnotations` thêm `annotationVersion: number`.
- `client/src/api.ts` — `saveAnnotations(imageId, annotations, expectedVersion?) → {annotations, annotationVersion}`.
- `client/src/pages/AnnotatorPage.tsx` — `annotationVersionRef` ref tracking; `scheduleSave` gửi version, catch 409, reload image.
- `tests/auth.test.js` — Row 18: 12 test case — conflict detection, backward compat, response shape. 113 passed, 0 failed.

Depth-1 callers của PUT annotations: `client/src/api.ts → AnnotatorPage.tsx` (`saveAnnotations`). Không caller server-side nào bị break (`history.js` không gọi PUT annotations route).

Watch out cho Phase 4/5: `annotations.js` PUT response đã thay đổi thành object — bất kỳ code nào gọi PUT và expect `Annotation[]` trực tiếp sẽ break. Hiện chỉ `AnnotatorPage` gọi và đã được cập nhật.

Phase 3.5 — Image done status — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-3.5):

- `server/src/db.js` — `m006_add_image_done_columns(): ALTER TABLE images ADD completed_at TEXT + completed_by INTEGER REFERENCES users(id) + index`; verify row count trước/sau (CTO condition #1 pattern).
- `server/src/routes/images.js` — 2 endpoint **MỚI:** `POST /:imageId/mark-done` (tất cả role, set `completed_at=now()/completed_by=req.user.id + logActivity`); `DELETE /:imageId/mark-done` (annotator chỉ bỏ của mình — 403 nếu người khác, reviewer/admin any — set NULL).

- `server/src/routes/reviews.js` — `submit-review` thêm `gate completed_at IS NOT NULL → 409 IMAGE_NOT_COMPLETED` nếu chưa done.
- `client/src/types.ts` — `ImageItem` thêm `completed_at?: string | null, completed_by?: number | null`.
- `client/src/api.ts` — thêm `markImageDone(projectId, imageId) + unmarkImageDone(projectId, imageId)`.
- `client/src/pages/AnnotatorPage.tsx` — `state doneBusy, handlers handleMarkDone/handleUnmarkDone`, phím tắt D (toggle done), nút "❑ Xong (D)" / "✓ Đã xong" trong toolbar.
- `client/src/pages/ProjectDetailPage.tsx` — `type DoneFilter`, filter dropdown "Đã hoàn thành / Chưa hoàn thành", badge "✓ Xong" màu xanh đậm trên image tile.
- `tests/auth.test.js` — Row 8 (submit-review): thêm mark-done trước mỗi submit; Row 19 (15 test case): gate 409, mark-done, unmark, role guard. 128 passed, 0 failed.

Depth-1 callers:

- `client/api.ts` → `AnnotatorPage.tsx` (`markImageDone/unmarkImageDone`)
- `reviews.js` `submit-review` gate → caller: `client/api.ts` → `AnnotatorPage.tsx` (`submitReview`)

Watch out cho Phase 4: Phase 4 (detect cache) không đụng bảng

`images/annotations/annotation_history` — không cần cập nhật done logic. Phase 5 (UX) cũng không đụng `completed_at/completed_by`.

Phase 4.1 — Detect cache — ✓ HOÀN THÀNH**Files đã thay đổi (STEP-4.1):**

- `server/src/db.js` — `m007_detect_cache()`: CREATE TABLE `detect_cache` (`id, image_id FK CASCADE, model_id FK CASCADE, raw_detections TEXT, created_at`) + UNIQUE idx (`image_id, model_id`) + 2 indexes; verify row count trước/sau (CTO condition #1 pattern).
- `server/src/routes/autolabel.js` — thêm `CACHE_RAW_CONF=0.01`; helpers `getCachedDetections()`, `saveToCacheDetections()`, `filterRawBoxesByConf()`; refactor `runInferenceHTTP()`: nhận thêm param `modelId`, tách hit/miss per batch, gọi inference với `CACHE_RAW_CONF` cho miss, lưu raw → filter → save; thêm `DELETE /cache` endpoint; cập nhật call site truyền `model_id`.
- `server/src/python/inference_service.py` — `_predict_single()`: thêm `conf` field vào mỗi box (`bbox: r.boxes.conf[i]`, `OBB: obb.conf[i]`) — backward compat (Node.js `saveDetectionsForImage` ignore field này).
- `tests/auth.test.js` — Row 20 (4 test case): `DELETE /cache` unauth→401, admin→200, annotator→200, bad project→404. Header comment cập nhật.

Verify row count (m007): CREATE TABLE mới — không có risk mất dữ liệu existing. Pattern giữ nhất quán với m005/m006.

Test kết quả: 133 passed, 0 failed, 0 skipped (128 cũ + 5 mới Row 20).

Depth-1 callers:

- `index.js` (`mount /api/projects/:pid/auto-label`) — không đổi mount path

- `client/api.ts startAutoLabel` — REST interface giữ nguyên (không cần sửa client)

Watch out cho STEP-4.2 (Prefill bbox):

- Dùng lại `getCachedDetections(imageId, modelId) + filterRawBoxesByConf (boxes, conf)` từ `autolabel.js` khi mở ảnh chưa label — 2 hàm này có thể import hoặc copy
- Cache key là `(image_id, model_id)` — STEP-4.2 cần biết `model_id` đang active của project
- `CACHE_RAW_CONF=0.01` đã được dùng để build cache; STEP-4.2 chỉ cần đọc cache + filter, không detect lại
- Nếu cache miss khi mở ảnh → STEP-4.2 tự quyết định: gọi inference ngầm hoặc hiện thông báo "chưa có cache"

Phase 5.1 — Undo/Redo

Files bị ảnh hưởng:

- `client/src/pages/AnnotatorPage.tsx` — thêm `undoStack/redoStack` state, `Ctrl+Z/Ctrl+Y` handler

Phase 6.1 — Model versioning

Files bị ảnh hưởng:

- `server/src/db.js` — `ALTER TABLE models ADD COLUMN metadata JSON`
- `server/src/routes/models.js` — thêm PATCH endpoint, GET endpoint với metadata
- `client/src/types.ts` — mở rộng `ModelInfo`

10. Lịch sử cập nhật

Ngày	Người cập nhật	Nội dung	Commit
---	---	---	---
2026-08-04	senior-developer (STEP-0.1)	Tạo mới — full audit 5 bảng, 8 routes, API endpoints, blast radius map	(STEP-0.1)
2026-08-04	senior-developer (STEP-1.1)	Cập nhật §3.10 <code>autolabel.js</code> (HTTP mode + legacy rollback), thêm §5.2 <code>inference_service.py</code> , cập nhật §8 Python deps, §9 Phase 1.1 DONE	(STEP-1.1)
2026-08-04	senior-developer (STEP-1.2)	Thêm §3.11 <code>jobs.js</code> (MỚI), cập nhật §2 (migrations/ + <code>jobs.js</code>), §3.1 (route <code>/api/jobs</code>), §3.2 (startup cleanup), §3.10 (DB-backed job storage), §6 (jobs table schema), §7 (<code>/api/jobs</code> endpoints), §9 Phase 1.2 DONE	(STEP-1.2)
2026-08-04	junior-developer (STEP-1.3)	Thêm §3.12 <code>thumbnails.js</code> (MỚI), cập nhật §1 (<code>data/thumbnails/</code>), §2 (thư mục + route), §3.1 (mount <code>/api/images</code>), §3.5 (thumbnail_url field), §4.3 (<code>ImageItem.thumbnail_url</code>), §7 (<code>/api/images/:id/thumb</code>), §9 Phase 1.3 DONE	(STEP-1.3)
2026-08-04	tech-lead+cto+em	Auth ADR riêng (<code>ADR-auth-labeling-</code>	(STEP-2.1)

	(STEP-2.1)	studio.md), security-audit-stride, CTO+EM APPROVED kèm điều kiện #A1/#A2	
2026-08-04	senior-developer (STEP-2.2)	Bảng users (INTEGER id, role, color), routes auth.js/users.js, middleware, LoginPage — 87 test pass ma trận AD-A5, npm audit sạch (điều kiện #A1/#A2 đạt), §9 Phase 2.2 DONE	(STEP-2.2)
2026-08-04	senior-developer (STEP-2.3)	Thêm §3.13 reviews.js (MỚI), cập nhật §6 bảng images (4 cột review), §9 Phase 2.3 DONE — review workflow submit/approve/reject, Row 8 test pass	(STEP-2.3)
2026-08-05	senior-developer (STEP-3.1)	Cập nhật §6 — cột annotations.version, bảng mới annotation_history (snapshot), activity_log; §9 Phase 3.1 DONE — verify row count trước/sau (0 mất dữ liệu, CTO condition #1 đạt)	(STEP-3.1)
2026-08-05	senior-developer (STEP-3.2)	Thêm §3.14 history.js (MỚI), cập nhật §2 (routes/history.js), §3.1 (mount /api/images/:id history), §3.6 (annotations.js + pruneAnnotationHistory), §7 (Annotation History endpoints), §9 Phase 3.2 DONE — 93 test pass (0 fail, 0 skip)	a4c3519
2026-08-05	junior-developer (STEP-3.3)	Thêm §3.15 activity.js (MỚI), cập nhật §2 (routes/activity.js), §3.1 (mount /api/projects/:id/activity + activityRouter), §3.2 (db.js exports: logActivity), §3.9 (images.js gọi logActivity), §3.10 (export.js gọi logActivity), §9 Phase 3.3 DONE — 101 test pass (0 fail, 0 skip)	a0f01dc
2026-08-05	senior-developer (STEP-3.4)	Cập nhật §3.5 (GET /:imageId trả thêm annotationVersion), §3.6 (annotations.js — optimistic locking: check expectedVersion trước transaction, PUT response đổi từ Annotation[] thành {annotations, annotationVersion}), §4.2 (api.ts — saveAnnotations thêm tham số expectedVersion), §4.3 (ImageWithAnnotations.annotationVersion), §4.4 (AnnotatorPage — annotationVersionRef, 409 handling), §7 (PUT annotations response 409), §9 Phase 3.4 DONE — 113 test pass (0 fail, 0 skip), tsc 0 lỗi	(STEP-3.4)
2026-08-05	senior-developer (STEP-3.5)	Cập nhật §6 (images — 2 cột mới completed_at/completed_by + index), §7 (Images — 2 endpoint MỚI POST/DELETE mark-done; reviews.js gate IMAGE_NOT_COMPLETED), §9 Phase 3.5 DONE — 128 test pass (0 fail, 0 skip), tsc 0 lỗi. Phase 3 HOÀN TOÀN HOÀN THÀNH.	(STEP-3.5)
2026-08-05	senior-developer (STEP-4.1)	Thêm bảng detect_cache (§6), route DELETE /cache (§3.10/§7), cache raw detections theo (image_id, model_id) trong	ac61d98→(STEP-4.1)

		<code>autolabel.js,</code> <code>inference_service.py</code> trả thêm <code>conf</code> , §9 Phase 4.1 DONE — 133 test pass (0 fail, 0 skip)	
--	--	--	--